# Saurabh Surashe

MTS III at Mavenir Systems

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## Professional Summary

C++ and Linux Systems Software Engineer with **4+ years of experience** developing high-performance, latency-sensitive networking software for carrier-grade wireless communication systems.

Experienced in Layer-2 scheduler architecture, protocol stack development, Linux systems programming, concurrent software, performance optimization and production debugging.

Designed networking software supporting **NB-IoT NTN**, delivering **production features, protocol enhancements, performance optimizations and customer fixes**.

Passionate about Networking Infrastructure, Distributed Systems, High Performance Computing, AI Infrastructure and Modern C++ software development.

## Core Competencies

**Programming Languages:** C++17, STL, C, Python, Java

**Linux & Systems:** Linux, POSIX, pthreads, Shared Memory IPC, Multithreading, GDB, Intel VTune, CMake

**Networking & Protocols:** TCP/IP, UDP, NB-IoT, LTE, Non-Terrestrial Networks (NTN), MAC, RRC, S1AP, Wireshark, ASN.1

**Software Engineering:** Concurrent Programming, Distributed Systems, Performance Optimization, Low-Latency Systems, Carrier-Grade Software

**AI / Machine Learning:** Python, Sentence Transformers, Semantic Search, Embeddings, TensorFlow, Keras

**Development Tools:** Git, Jenkins, Jira

## Professional Experience

**Mavenir Systems Pvt. Ltd.** — July 2022 – Present

*Member of Technical Staff I → Member of Technical Staff III* — *Bengaluru, India*

- Designed and developed high-performance Layer-2 and Layer-3 networking software in C/C++ for a carrier-grade NB-IoT wireless communication platform supporting both terrestrial and Non-Terrestrial Networks (NTN).
- Redesigned the latency-sensitive MAC scheduler resource allocation workflow, improving uplink throughput by approximately **40%** while preserving deterministic scheduling deadlines for real-time deployments.
- Architected key Layer-2 scheduler enhancements for **3GPP Release-17 Non-Terrestrial Networks (NTN)**, implementing Timing Advance computation, Doppler compensation, dynamic satellite ephemeris management and interlaced scheduling for GEO and LEO satellite communication.
- Architected a low-latency **Shared Memory IPC** framework enabling runtime propagation of satellite ephemeris information across distributed Layer-2 software components without service interruption.
- Implemented more than **60 production KPIs** providing detailed visibility into scheduler behaviour, protocol execution, resource utilization and overall software health, substantially improving runtime observability and customer debugging.
- Profiled latency-sensitive software using **Intel VTune**, Linux perf and GDB to identify CPU bottlenecks, optimize execution paths and improve runtime performance.
- Performed systematic root-cause analysis using protocol traces, Wireshark captures, runtime logs, execution profiling and live debugging to investigate and resolve complex customer-reported production issues.
- Collaborated closely with Layer-1, Layer-2, Layer-3, Platform, QA, DevOps and Customer Engineering teams throughout architecture, implementation, validation and commercial deployment.
- Mentored two junior engineers through design discussions, debugging sessions, code reviews and knowledge sharing while promoting engineering best practices.

## Projects

**AI-Powered Job Discovery & Matching Platform** | *Python, FastAPI, SQL, Sentence Transformers, SMTP*

- Developed a modular Python platform automating job discovery and semantic matching using the `all-MiniLM-L6-v2` transformer model to calculate resume-to-job description similarity scores.
- Built asynchronous scraping pipelines using `asyncio` to concurrently fetch and parse opportunities from major ATS engines including Workday, Greenhouse, Eightfold, and SmartRecruiters.
- Implemented highly optimized, in-memory semantic similarity search utilizing NumPy vector calculations, achieving sub-200ms scoring latency without database-side vector extensions.
- Designed clean REST APIs with FastAPI to support resume parsing, job indexing, user subscription profiles, and score-threshold configurations.
- Integrated an automated notification system sending formatted HTML job digests via SMTP for jobs matching custom score thresholds, deployed via serverless pipelines and background tasks.

## Achievements

**Mavenir Spotlight Award** (Received Multiple Times) for delivering high-impact production features, technical excellence and successful customer software releases.

Co-authored: **"Privacy and Security Comparison of Web Browsers: A Review"**, AINA 2022 (*SpringerLink*)

Co-authored: **"Enhancing Cyber Security: Malicious web page detection using Machine and Deep Learning"**, AINA 2025 (*SpringerLink*)

## Education

**National Institute of Technology Surathkal** — Karnataka
*Master of Technology in Computational Data Science* — *June, 2020 – May, 2022*

**MIT College of Engineering Pune** — Maharashtra
*Bachelor of Engineering in Mechanical Engineering* — *June, 2015 – May, 2019*